THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Heat flows from a body at higher temperature to a body at: 1

(a) lower temperature **(b)** absolute zero only
(c) higher temperature **(d)** same temperature only

2. In conduction, the particles of the solid: 1

(a) pass on heat to neighbours but stay in position
(b) move over long distances
(c) fly off
(d) stop

3. Dark-coloured clothes absorb ___ heat than light-coloured clothes. 1

(a) more **(b)** equal **(c)** no **(d)** less

4. The three modes of heat transfer are: 1

(a) radiation, freezing, boiling **(b)** conduction, convection, radiation
(c) convection, sublimation, condensation **(d)** conduction, evaporation, melting

5. In convection, heat is carried by: 1

(a) moving particles of the fluid **(b)** vibrating solid
(c) waves in space **(d)** light only

6. In summer we prefer to wear: 1

(a) dark-coloured clothes **(b)** woollen clothes
(c) light-coloured clothes **(d)** leather clothes

7. Transfer of heat through a solid without movement of the solid is: 1

(a) radiation **(b)** evaporation **(c)** conduction **(d)** convection

8. Which of these needs no material medium? 1

(a) convection **(b)** radiation **(c)** conduction **(d)** all three

9. In winter we prefer to wear: 1

(a) woollen clothes **(b)** thin cotton **(c)** net clothes **(d)** plastic

10. Transfer of heat by actual movement of a fluid is: 1

(a) radiation **(b)** conduction **(c)** convection **(d)** insulation

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Name the three modes of heat transfer. 2

12. Why do we wear light-coloured clothes in summer? 2

13. Why are cooking utensils made of metal? 2

14. What is conduction? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to show conduction in a metal rod. 3

16. Why are heaters placed near the floor and coolers/ACs near the ceiling? 3

17. Why do birds fluff their feathers in cold weather? 3

OR

Explain how convection currents are set up in water heated in a beaker. 3

18. Explain why a metal feels colder than wood at the same temperature. 3

OR

Why do we use a cloth to hold a hot vessel? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain conduction, convection and radiation with one activity or example each. 5

OR

Differentiate between conduction, convection and radiation on five points. 5

20. Describe the experiment to show convection in water using potassium permanganate crystals. 5

OR

Explain the role of conduction and convection in the cooking of food in a pan. 5

END OF QUESTION PAPER — Total: 40 Marks

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